



Structure-Property Relationships in Cellulose-Based Hydrogels

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Abstract

Hydrogels are widely used for different biomedical applications, due to their ability to absorb, retain, and release water solutions in a reversible manner and in response to specific environmental stimuli. The review is focused on the preparation methods, main characteristics, as well as biomedical applications of hydrogels prepared from the most abundant biopolymers on earth, cellulose. The chapter emphasizes the latest developments in the design and manufacture of cellulose-based hydrogels. The preparation of hydrogels without covalent cross-links (physical hydrogels) and with covalent cross-links (chemical hydrogels) is discussed. The behavior of gels upon coagulation and the swelling

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capacity in water were analyzed. A systematic investigation into the structure and physical-chemical properties of cellulose-based hydrogels was performed in order to describe the relationships between the network structure and gel properties. The degree of cross-linking of the hydrogels, the morphology of the three-dimensional (3D) matrices, the bulk geometry, and the description by different complementary techniques which offered insight into structure-property relationships of hydrogels are reviewed. The sorption properties of cellulose-based hydrogels and the effect of the design parameters of hydrogels on their biomedical applications are also discussed.

Keywords

Cellulose · Hydrogel · Gelation · Cross-linking · Swelling · Drying · Morphology

1 Introduction

Hydrogels are defined as a hydrophilic polymers structure, capable to absorb and retain a relatively large amount of water or aqueous fluids (up to 10 g/g for ordinary absorbent hydrogels or more for superabsorbent hydrogels). The water-absorbing capacity and mechanical property of hydrogels usually rely on their three-dimensional (3D) network structure which can be driven by both chemical and physical interactions [1–3].

The classification of hydrogels depends on their physical properties, nature of swelling, method of preparation, origin, ionic charges, sources, rate of biodegradation, and the nature of cross-linking [4]. A detailed classification of hydrogels is presented in Fig. 1.

On the basis of the preparation method, the hydrogels can be divided into (i) *physical hydrogels*, which are achieved via physical processes, such as hydrophobic association, chain aggregation, crystallization, polymer chain complexion, and hydrogen bonding, and (ii) *chemical hydrogels*, formed by chemical covalent cross-linking. Physical hydrogels are reversible due to the conformational changes, while chemical hydrogels are permanent and irreversible because of configurationally changes [4]. Concerning the source, hydrogels can be divided into those obtained from natural polymers, synthetic polymers, and the hybrid hydrogels. Over the past two decades, natural polymers have been slowly replaced by synthetic ones or were hybridized to provide a higher service life, gel strength, and water absorption capacity [5].

The physicochemical properties of hydrogels are based on a number of factors, including the chemical nature of the monomer, the method of polymer synthesis, their molecular weight due to the reaction environment, and macromolecular structure. On the other hand, the physical structure of the polymer relies on the strength of the covalent bonds, the rigidity, and intermolecular force strength within the polymer chain [6].

The volume or mass-swelling ratio of the hydrogel is the most important variable to be evaluated for given environmental conditions, as it affects the diffusive,

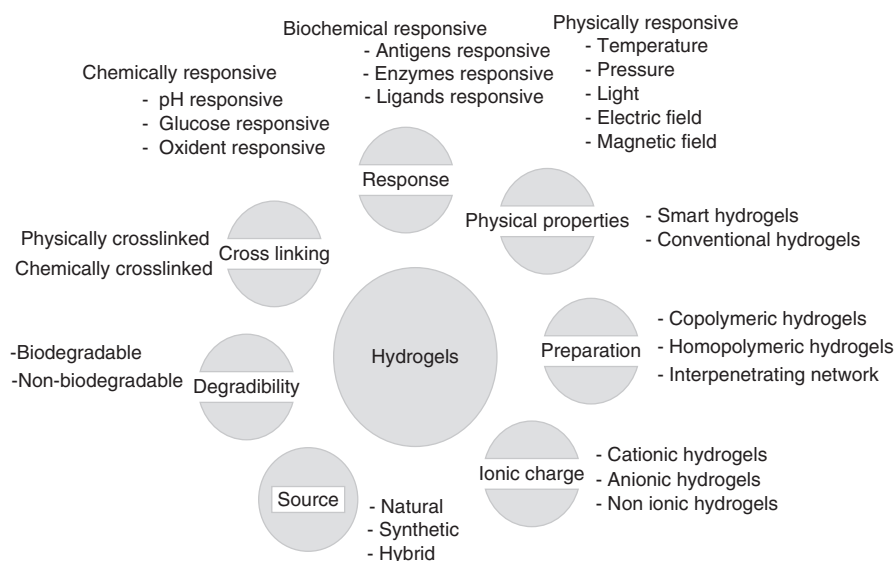


Fig. 1 Classification of hydrogels based on the different properties. (Reprinted with permission from [4]. Copyright © 2015, Elsevier.)

mechanical, optical, acoustic, and surface properties of the hydrogel itself. The amount of water retained by the mesh of the hydrogel network depends on the structure of the polymer network itself and on the environmental conditions, such as the temperature, pH, and ionic strength of the water solution in contact with the polymer [7].

The interactions responsible for the water sorption include capillary, osmotic, and hydration forces, which are counterbalanced by the forces exerted by the cross-linked polymer chains in resisting expansion. The equilibrium swollen state depends on the magnitudes of these opposing effects and determines to a large extent some important properties of the hydrogel, including internal transport and diffusion characteristics and mechanical strength. Many of these properties are governed not only by the degree of swelling, but also directly by the chemical nature of the polymer network and the network morphology [8].

The physical or chemical processes are adjustable based on the desired characteristics, such as structure, matrix density mechanical strength, biocompatibility, and biodegradability, which are subjective to the specific field and function to which they will be applied [9].

Since environmentally friendly products and processes are of great importance nowadays, much industrial research has focused attention on the synthesis of novel cellulose-based hydrogels. In fact, cellulose is a natural polymer that can be employed as renewable-based polymeric material with biodegradable properties [7].

The hydrogels from natural polymers, especially those from cellulose, are promising for their application in biomaterial field, because of their unique advantages,

such as abundance, non-toxicity, biocompatibility, biodegradability, and biological functions [10–12].

Innovative cellulose-based hydrogels have been developed either as water absorbents for specific applications, including personal hygiene products, underwater devices, body water retainers [7, 13], or biomedical devices, as ocular bandages [14], artificial cartilage [15, 16], controlled drug delivery [17–19], scaffolds for regenerative medicine [20, 21], stomach bulking agents [22], and wound dressings [12, 23, 24].

This chapter explores the synthesis and design of cellulose-based hydrogels, including pure cellulose and cellulose composite. The recent developments with emphasis on the fabrication, properties, and possible applications are highlighted. In particular, the attention is focused on the structure-property relationships of hydrogels and their biomedical and pharmaceutical applications.

2 Cellulose and Its Applications

Cellulose is the most common organic polymer and is considered an almost inexhaustible source of raw material for the increasing demand of environmentally friendly and biocompatible products [25]. This biopolymer exhibits some excellent properties, such as mechanical robustness, biodegradability, hydrophilicity and biocompatibility, properties responsible for the extensive use of cellulose in a wide spectrum of applications that include pharmacy, agriculture, medical science, industries, and many other related branches [26].

In contrast with its vast potential of applications, it is known that there are certain difficulties in producing cellulose-based products such as films, fibers, membranes, hydrogels, and beads, due to its partially crystalline structure and strong hydrogen bonds. The long-chain cellulose polymers consist of D-glucose subunits which are linked together by β -1,4 glycosidic bonds. These linear polymers are linked together by different inter- and intramolecular bonds, which allow them to be packed side by side in planar sheet and bundled into microfibrils (Fig. 2). Hence, the cellulose is insoluble in water as the hydroxyl groups in sugar chains are bonded to each other, making a hydrophobic scenario [27].

Unlike other types of water-soluble polysaccharides, cellulose requires separate solvent systems for its dissolution, which have remained limited because of the lack of a suitable solvent. In the early days, the use of concentrated sodium hydroxide (NaOH) solution as cellulose solvent interested researchers due to numerous applications of cellulose treatments with NaOH solution in fiber modification, dissolution, and regeneration [28]. Moreover, there were other solvent systems, from which the most widely used until the 1950s was cuprammonium hydroxide (Cuam) solution, while the aqueous solutions of cupriethylenediamine (Cuem) complex and some tetraalkylammonium hydroxides were used as solvents for analytical purposes in those days. In the decade of the 1960s, the discovery of numerous metal-complex solvents brought in more variations of cellulose solvents from which

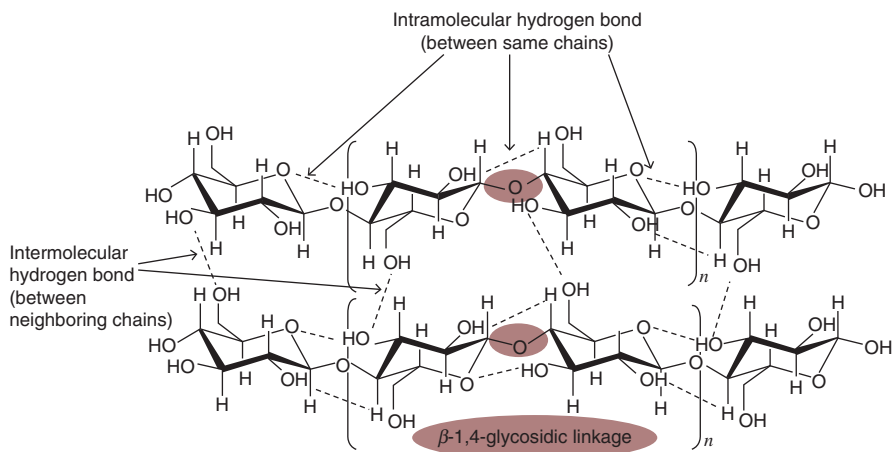


Fig. 2 Chemical structures of cellulose chains. (Reprinted with permission from [27]. Copyright © 2014, Hindawi.)

the most important were ferric sodium tartrate (FeTNa) and cadoxen solvent systems.

In recent years, considerable attention has been directed to several cellulose-solvent systems for dissolving cellulose, such as *N,N*-dimethylacetamide/lithium chloride (DMAc/LiCl) systems [29, 30], *N*-methylmorpholine-*N*-oxide (NMMO) [31, 32], tetrabutylammonium fluoride/dimethylsulfoxide (TBAF/DMSO) [33], paraformaldehyde/dimethylsulfoxide (PF/DMSO), triethylammonium chloride/dimethylsulfoxide (TEAC/DMSO) [34], ionic liquids [35–37], and alkali aqueous solutions [38–41].

In spite of its poor solubility characteristics, cellulose has versatile uses in many industries such as veterinary foods, wood and paper, fibers and clothes, cosmetic, and pharmaceutical industries [42, 43].

Generally, cellulose is strong, low cost, reproducible, recyclable, and biocompatible [44]. Thanks to all of these properties, they are often used in various biomedical applications such as blood purification membranes in artificial kidneys [45], membranes in plasmapheresis [46], body water retainers [13], ocular bandages [14], artificial cartilage [15], scaffolds for regenerative medicine [47], and stomach bulking agents [48].

3 Synthesis of Cellulose II-Based Hydrogels

Cellulose hydrogels are a special class of cellulose materials which can be based on:

- **Cellulose I**, case of bacterial cellulose (BC) or cellulose nano-/microfibrils (CNF) or cellulose nanocrystals (CNC) [49, 50]
- **Cellulose II**, prepared via cellulose dissolution-coagulation route [51, 52]

Both types show high water retention, from several hundred to several thousand percent.

Cellulose-based hydrogels can be obtained via either physical or chemical stabilization of aqueous solutions of cellulose. Additional natural and/or synthetic polymers might be combined with cellulose to obtain composite hydrogels with specific properties [7]. The synthesis of cellulose-based hydrogels generally consists of two steps [53]: (i) dissolution of cellulose fibers or powder, and (ii) chemical and/or physical cross-linking, in order to obtain a three-dimensional network of hydrophilic polymer chains, which is able to absorb and retain a significant amount of water. By tuning the cross-linker and cellulose concentrations, it is possible to optimize the hydrogel mechanical properties and swelling capabilities.

3.1 Dissolution of Native Cellulose

Due to its highly extended hydrogen-bonded structure, cellulose is very difficult to dissolve in common solvents, and thus only few solvents have been used for its solubilization.

A series of solvent systems such as LiCl/DMAc for cellulose powder, paraformaldehyde (PF)/dimethyl sulfoxide (DMSO), and triethylammonium chloride (TEAC)/DMSO for cellulose pulp have been used to fabricate cellulose hydrogels [52]. The obtained physical properties of hydrogels, such as transparency, strength, and water content, mainly depend on the composition of the coagulation/regeneration bath, while in the case of a nonaqueous solution, the maximum transparency of the hydrogels was obtained. A novel method, deionization of the cellulose/LiCl/DMAc solution with ion exchange resins, has been reported for preparation of gel beads that appear colorless and transparent without a fibrous texture compared with the “water-coagulated gel,” probably because the molecular association (crystallite formation) precedes the phase separation during ion exchange [54].

By using N-methylmorpholine-N-oxide (NMMO), with low toxicity, high thermal stability, and the possibility of being recovered and reused, an opalescent cellulose-based hydrogel was obtained by adding excess water into the cellulose/NMMO/water solution [55].

Another solvent system for cellulose dissolution and gelation is ionic liquids (ILs). The preparation of the clear and homogeneous cellulose solution was done under mild conditions by using N-ethyl-N'-methylimidazolium methylphosphonate ($[\text{C}_2\text{mim}][(\text{MeO})(\text{H})\text{PO}_2]$) [56]. The mechanical strength of cellulose hydrogels strongly depends on the concentration and degree of polymerization (DP) of cellulose. Moreover, the mechanical property of cellulose gel remained unaltered after treatment with boiled water, indicating the excellent thermal stability of cellulose hydrogels. Various ILs used for cellulose dissolution have been applied in the preparation of cellulose-based hydrogels, such as 1-ethyl-3-methylimidazolium acetate ($[\text{C}_2\text{mim}][\text{OAc}]$) [57, 58], 1-butyl-3-methylimidazolium chloride ($[\text{C}_4\text{mim}]\text{Cl}$), and 1-allyl-3-methylimidazolium chloride ($[\text{Amim}]\text{Cl}$) [54].

A recent alternative for cellulose solubilization and hydrogel preparation is the alkali aqueous system, which uses freezing at low temperature followed by the thawing of the mixture, instead of stirring at room temperature or high temperatures. The obtained physically cross-linked hydrogels have a dense texture with heterogeneous morphology, which presents pores between 4 and 8 μm [59]. The bulk density for the physical cross-linked hydrogel is twice higher than for chemical cross-linked hydrogel. An interesting fact is that the physical cross-linked hydrogels are opaque, while chemical cross-linked hydrogels become more and more transparent, by increasing the cross-linker concentration (epichlorohydrin).

The cellulose dissolution at low temperature arises as a fast dynamic self-assembly process among solvent small molecules (NaOH, urea, and water) and the cellulose macromolecules [38]. Cellulose in solution behaves as random coils, semiflexible chains, or entangled chains, and the degree of entanglement depends on the polymer concentration. The gel structure becomes more and more organized and stable as cellulose concentration increases due to the higher degree of entanglements and the more hydrogen-bonding interactions that exist. Cellulose hydrogels can be formed via destructing the stability of cellulose solution by increasing the temperature to 50 °C or reducing it to −20 °C [52]. When a low temperature was used, active NaOH hydrates were bound to cellulose to bring it into the solvent, whereas urea hydrates served as the hydrogen-bonding donor and receptor to prevent the approach toward each other of the cellulose molecules, leading to a good dispersion of cellulose in the solution. At elevated temperature the NaOH and urea hydrates bounded on the cellulose chains were perturbed, so the junction between cellulose molecules occurred as a result of the self-association force of cellulose, resulting in the formation of a network structure [60].

3.2 Physical and Chemical Gelation

A key step in the formation of any hydrogel is “gelation,” which can be interpreted as the interconnecting of the macromolecular chains in some manner, such that essentially the whole structure becomes linked together [61, 62].

Cellulose in solution behaves as random coils, semiflexible (or semirigid) chains, or entangled chains, and the degree of entanglement depends on the polymer. While polymer solutions of low concentrations are completely isotropic, with increasing polymer concentration, the transition from solution to a liquid crystalline gel takes place followed by gelation into a solid gel that has an anisotropic structure [54]. The gel structure becomes more and more organized and stable as biopolymer concentration increases due to the higher degree of entanglements and the more hydrogen-bonding interactions that exist [58].

The gelation mechanism involves hydrophobic associations among the macromolecules possessing the methoxy group. At low temperatures, polymer chains in solution are hydrated and simply entangled with one another. As temperature increases, macromolecules gradually lose their water of hydration, until polymer-

polymer hydrophobic associations take place, thus forming the hydrogel network [51].

The stability of the hydrogel can be enhanced by coagulation in different anti-solvents, as water [57], ethanol [63], methanol [64], $\text{H}_2\text{SO}_4/\text{Na}_2\text{SO}_4$ [65], or by curing (i.e., keeping the solution at various temperatures between 5 °C and 60 °C) [66].

Covalent cross-linking has emerged as the preferred means of converting suitable hydrophilic polymers into absorbent hydrogels, and the cross-linking optimization implies a balance between achieving high absorbency, suitable mechanical strength, and minimizing the amount of material that dissolves into the bulk of the liquid when a highly absorbent hydrogel is in use [61, 67, 68]. Stable and stiff networks with irreversible cross-linking bonds between the cellulosic chains can be obtained by chemical cross-linking reaction in the presence of chemical agents (epichlorohydrin, aldehydes and aldehyde-based reagents, urea derivatives, carbodiimides, and multi-functional carboxylic acids) or physical treatments (i.e., high-energy radiation) [51].

Intense research has focused on the preparation of cellulose-based hydrogels via cellulose dissolution/coagulation route, with continuous improvements of efficiency in solvents such as NaOH aqueous solutions and similar systems (i.e., NaOH/urea, NaOH/urea/thiourea, LiOH/urea) [69–72], which are non-derivatizing, inexpensive, and nontoxic cellulose solvents. In the presence of 7–9 wt% NaOH-water solvent, cellulose solutions undergo a physical gelation with the increase of time and temperature [73].

Gelation occurs due to cellulose chains self-association because of the preferential cellulose-cellulose and not cellulose-solvent interactions; this physical gelation is accompanied by a micro-phase separation [59]. By adding epichlorohydrin (ECH) as cross-linker in cellulose-NaOH-water solution, cellulose chains become chemically cross-linked. Two mechanisms are therefore present and can possibly compete: physical gelation due to chains self-association and chemical cross-linking (Fig. 3a).

It was suggested that chemical bonds act as “spacers” preventing, at least partly, the self-association of cellulose chains, as shown on a sketch in Fig. 3a. A schematic presentation of the structure of physical and chemical cellulose gels is shown in Fig. 3b. Chemical gelation (i) perturbs cellulose chains self-association and packing which leads to the decrease of crystallinity, (ii) leads to a more homogeneous morphology which results in transparent swollen coagulated cellulose hydrogels, and (iii) increases swelling in water and adsorption of water vapors due to a more porous structure. As a consequence, the difference in the structure and swelling of physical vs. chemically cross-linked cellulose influences their release properties: in chemically cross-linked cellulose, a larger amount of drug can be loaded, and the release kinetics is faster compared to physically cross-linked matrix. By varying cellulose concentration and the amount of cross-linker, it is possible to prepare versatile cellulose hydrogels and dry porous networks with controlled morphology and porosity.

Compared to chemically cross-linked hydrogels which require covalent bonding for the formation of 3D networks, physically cellulose-based hydrogels are formed mostly by hydrogen bonding via a self-assembly process.

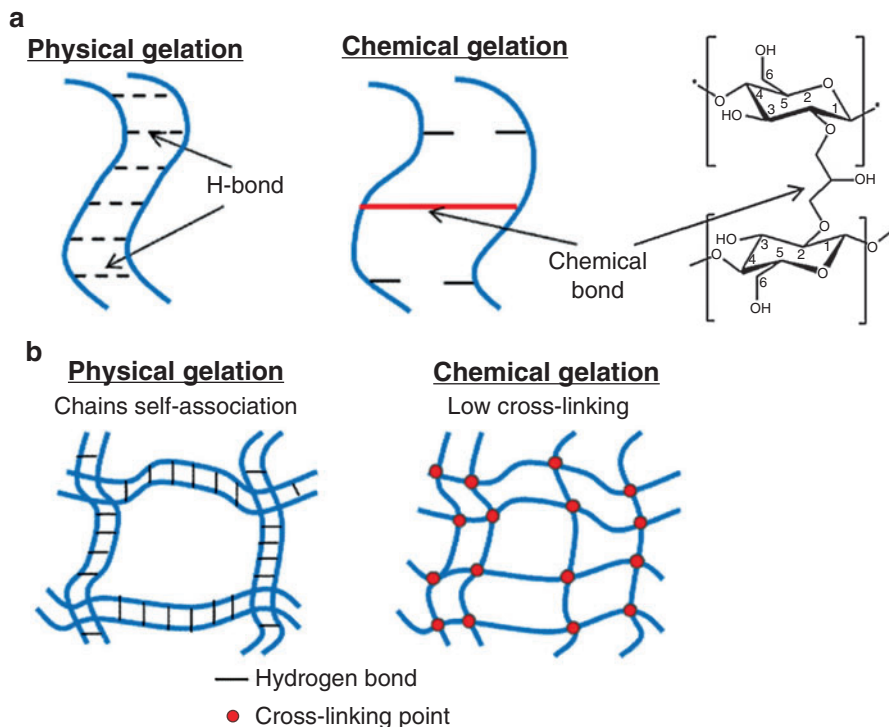


Fig. 3 (a) A sketch of network formation in cellulose solutions: physical gelation via self-association of chains and chemical cross-linking, (b) a schematic presentation of the structures of physical and chemical cellulose gels. (Reprinted with permission from [59]. Copyright © 2016, Elsevier.)

In the presence of strong organic solvents such as *N,N*-dimethylacetamide (DMAc)/LiCl, the gradual dissolution of cellulose starts by the formation of hydrogen bonding between the hydroxyl protons of the anhydroglucose unit within cellulose with the Cl^- in LiCl. The Cl^- is then attracted to the $(\text{LiDMAc})^+$ macrocation with the formation of repulsive forces between different segments of the host cellulose. The repulsive forces will then open more micropores, allowing more solvent to penetrate into the cellulose matrix. Due to the anhydrous nature of the above solution, the presence of moisture leads to the regeneration of hydrogen bonding between the dissolved cellulose segments, that is, gelation [62]. Throughout this process, the cellulose main chains assemble into a 3D network structure by reforming hydrogen bonds between various OH^- groups on the anhydroglucose units. Due to the fast reaction kinetics and high hydrophilicity of the anhydroglucose units, large amounts of moisture are trapped in the gel. Since the water molecules are present mostly between the anhydroglucose chains, the drying of hydrogels will lead to the formation of new hydrogen bonds and cause the collapse of micropores.

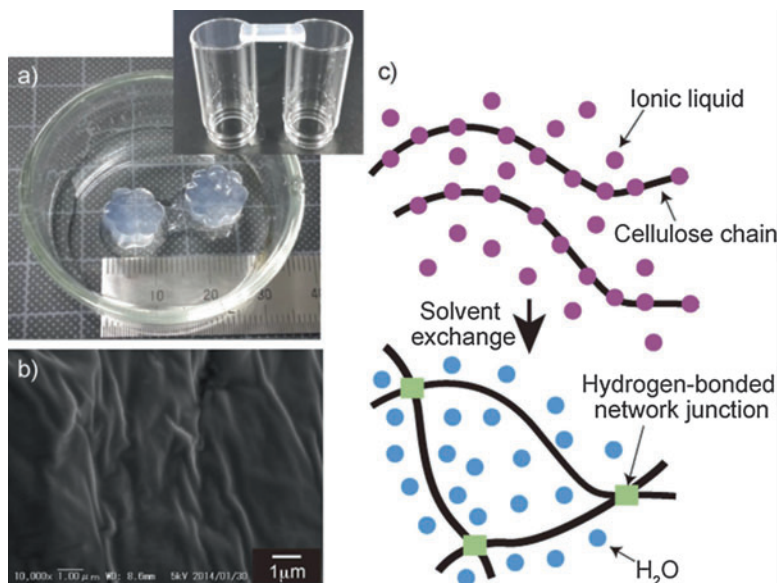


Fig. 4 (a) Optical images of flower-shaped cellulose hydrogels prepared from 5 wt% IL solution of WP. The inset is an optical image of bridged cellulose hydrogel. (b) SEM image of the surface of dried cellulose hydrogel. (c) Schematic gelation process of the cellulose solution. (Reprinted with permission from [37]. Copyright © 2016, John Wiley & Sons, Inc.)

Ionic liquids enable the dissolution of cellulose by disrupting and breaking hydrogen-bonding networks. By using N-ethyl-N'-methylimidazolium methylphosphonate ($[\text{C}_2\text{mim}][(\text{MeO})(\text{H})\text{PO}_2]$), a 10 wt% homogeneous cellulose solution, under mild conditions, has been prepared [56, 74].

Recently, cellulose-based hydrogels were prepared through the physical aggregation of polymer chains determined by hydrogen bonds, crystallization, helix formation, and complexation (Fig. 4). Cellulose was completely dissolved in $[\text{C}_2\text{mim}][(\text{MeO})(\text{H})\text{PO}_2]$ through the breaking of the hydrogen bonds among cellulose backbones, and the dissociated cellulose chains were partially aggregated during the solvent exchange process by treating them with methanol vapor. The structure was set uniformly by a spinodal decomposition-type phase separation during the treatment with methanol vapor, and the partial aggregation of cellulose chains acted as a network junction point in three-dimensional networks [37].

Cellulose hydrogels containing 95–99% water have been fabricated using the solvent exchange of homogeneous cellulose solutions without the use of chemical cross-linking (Fig. 4c). Water was incorporated within the three-dimensional network of the hydrophilic cellulose chains containing hydrogen-bonded network junction points.

4 Synthesis of Cellulose I-Based Hydrogels

4.1 Nanocellulose Hydrogels

Nanocelluloses are a class of natural materials derived from cellulose, the most abundant renewable polymer on earth, which include cellulose nanofibrils (CNFs) and cellulose nanocrystals (CNCs). Cellulose nano- and microfibrils are substructural elements of cellulosic fibers that can be mechanically disintegrated (high-pressure homogenization and refining, microfluidization, grinding, cryocrushing, and high-intensity ultrasonication combined with chemical or enzymatic treatments) from the cell wall matrix, while cellulose nanocrystals are isolated from microcrystalline cellulose using acid hydrolysis and/or sonication process.

The most crystalline form of cellulose is cellulose nanocrystals (CNCs), which are shorter than the nanofibrils. In addition, CNCs present many advantages, such as nanoscale dimension, high specific strength and modulus, high surface area, unique optical properties, and therefore wide possibilities of application [75]. Cellulose sources are variable and their degree of crystallinity strongly influences the dimensions of the liberated crystals [49, 76].

The physical and chemical properties of nanocelluloses have enabled their use in a wide variety of hydrophilic and hydrophobic composite matrices and hybrid materials, including hydrogels and aerogels. Furthermore, their high surface area-to-volume ratio enables enhanced interactions with, and binding to, polymers, other nanoparticles, and small molecules [77].

Hydrogels prepared only from CNCs at a concentration above 10 wt% will enter an aggregated gel-like phase [78, 79]. Other methods to prepare CNC hydrogels are by destabilizing the CNC suspensions with salt or acid, or by cross-linking of CNCs with multivalent ions or chemical bonds [80, 81]. In practice, the CNC concentration at which a gel is formed may be manipulated via changing the solution conditions, surface modifications, and/or adding adsorbing or non-adsorbing water-soluble polymers [82, 83]. It was observed that by increasing the ionic strength of CNC suspensions via salt addition, the electrostatic repulsion was suppressed, which led to dominant attractive forces (e.g., van der Waals and hydrogen bonding) and, consequently, to gelation. Thus, the increasing of the charge number and the radii of the ions both led to increases in gel stiffness [79, 81].

Due to their increased flexibility and tendency for entanglement, cellulose nanofibrils are more liable to hydrogel formation than CNCs.

The NFC suspensions (Fig. 5) bear the appearance of highly viscous shear-thinning transparent gels and have high aspect ratios and specific surface areas combined with remarkable strength and flexibility, low thermal expansion, high optical transparency, and specific barrier properties [84].

Typical CNF hydrogels contain 0.05–6 wt% CNFs, but CNF gels have been reported at concentrations of 0.125 wt%, following an enzymatic treatment and homogenization of bleached sulfite pulp [85]. These hydrogels are of two orders of magnitude concentration lower than that required for CNC gelation.

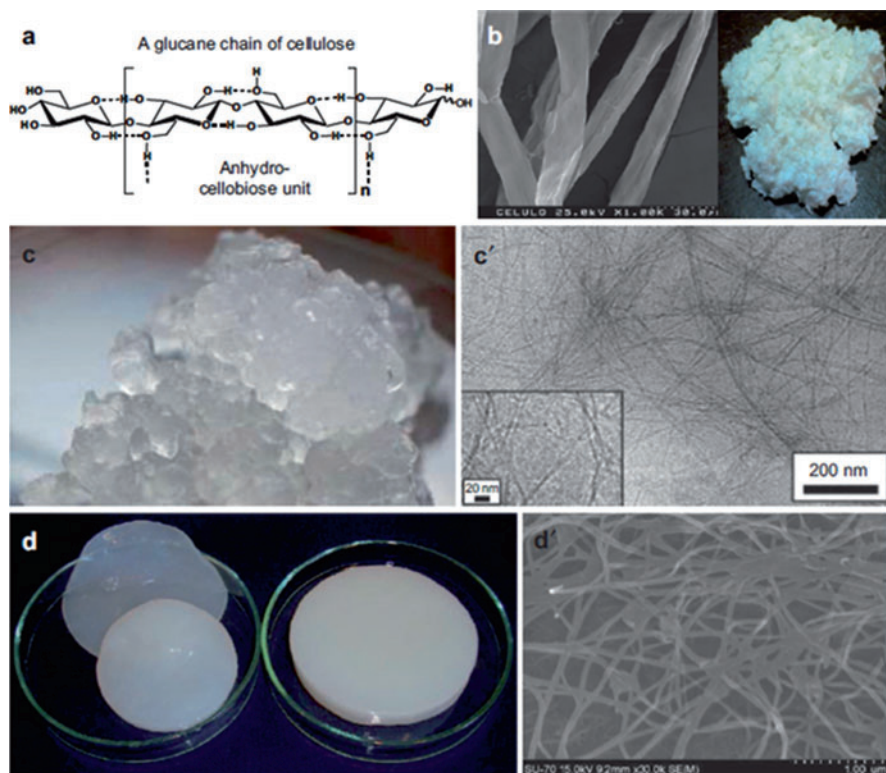


Fig. 5 Images of cellulose. (a) A glucan chain of cellulose with repeating anhydrocellobiose units. (b) Macroscopic and SEM images of conventional pulp fibers. (c and c') Macroscopic and SEM images of NFC. (d) Image of BC. (d') SEM image of BC. (Reprinted with permission from [84]. Copyright © 2011–2017, Walter de Gruyter GmbH.)

Hydrogels with oriented fibrils can be prepared by flow focusing CNF suspensions, followed by inducing gelation through pH reduction or salt addition, and these can be used as templates for anisotropic nanocomposites (Fig. 6) [79, 86, 87].

The flow-induced CNF-oriented alignment displayed exceptional mechanical properties that were enhanced as the degree of orientation was increased; however, filament stability drastically decreased in the presence of water, representing a key limitation of this technique [88]. Similar to most CNC hydrogels, CNF hydrogels lack macroscopic structure or hierarchy, which is a serious disadvantage when developing biomimetic materials or hydrogels for many biomedical applications [79].

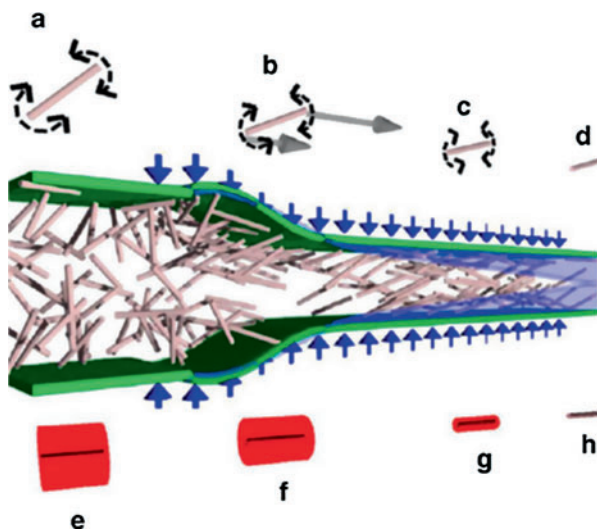


Fig. 6 Illustration of the flow focusing CNF hydrogel assembly process. The diffusion of Na^+ ions into the flow channel is depicted by blue arrows, the concentration of which increases as the channel narrows, shown by the shaded blue area. (a) Initially, CNFs undergo Brownian diffusion, which decreases as (b) individual particles undergo hydrodynamically induced alignment. (c) Particles are constrained uniaxially and (d) “frozen” in place as the increased Na^+ ions cause gelation of the system. (e–h) The electrostatic repulsion and the Debye length of CNF fibrils (shown in red) decrease along the channel as surface carboxyl groups become uncharged/shielded. (Reprinted with permission from [79]. Copyright © 2017, American Chemical Society.)

4.2 Bacterial Cellulose Hydrogels

Bacterial cellulose (BC), also known as microbial cellulose, is produced by different bacteria genera, such as *Gluconacetobacter*, *Sarcina*, and *Agrobacterium*, but *Gluconacetobacter xylinus* is probably the most commonly referred strain in this context [84]. A consideration of the nanocrystallinity of cellulose leads directly to BC, because this is synthesized in highly crystalline forms and in very high yields by *Gluconacetobacter xylinus* (*G. xylinus*), which is the most widely studied source of BC [89, 90]. BC can be produced as a highly swollen hydrogel and, depending on the static or agitated nature of the culture media, as a membrane (Fig. 5d, d') or in the form of small beads. More precisely, BC is a 3D network consisting of nano- and microfibrils with the dimensions of 3–4 nm thickness and 70–80 nm length (Fig. 5d'); that is, the fibrils are approximately 1000 times thinner than typical plant cellulose fibrils. These dimensions explain the unique properties of BC. Additionally, BC is free of lignin, hemicelluloses, and other natural components usually associated with cellulose isolated from the cell wall of plants [84]. BC has been shown to have superior properties compared to plant-derived cellulose, such as high surface area, degree of polymerization, wet tensile strength, purity, crystallinity,

and nanostructured fibers [91]. BC is known to be biocompatible, perhaps because of its flexibility and porosity, resembling properties of collagen in the human body [92].

A detailed analysis of the water interaction properties in as-biosynthesized BC gels indicates that only about 10% of the 99 wt% water present in BC gels behaved like free bulk water, and the majority of the water molecules in the gels are more or less tightly bound to the cellulose [93]. In a recent study, BC nanocellulose was shown to effectively control the uptake and release of albumin and demonstrated the retention of protein integrity, which can sometimes be compromised during the manufacturing process [94].

On the basis of the high purity and unusual physicochemical properties, BC-based hydrogels offer a wide range of applications in biomaterials fields such as tissue engineering scaffold [95], meniscus implant [96], and dental implants [97].

5 Multipolymer Hydrogels Based on Cellulose

The multipolymer hydrogels have been prepared from cellulose and other polymers by interpenetrating polymer networks (IPNs) technology, in order to combine the different properties of cellulose and other polymers. Generally, the preparation processes include the dispersion or dissolution of two or more polymers (usually using the same solvent) and the cross-linking step.

Cellulose can be blended with *natural biodegradable polymers*, such as chitin [98], chitosan [99], starch [100], xanthan [101], lignin [102], and alginates [103].

The blending of cellulose and chitin in the same solvent system is an effective way to produce phase-separated composite hydrogels. Chitin/cellulose beads have been prepared by coagulating a mixture of 4 wt% cellulose solution and 2 wt% chitin solution in 6 wt% NaOH/5 wt% thiourea aqueous system [52]. The adsorption ability of obtained beads is higher than that of pure chitin flakes, owing to the relatively large surface area and their high hydrophilicity. The hydrophilic skeleton and microporous-network structure of cellulose improve the adsorption ability of chitin [104].

Cellulose/chitosan hydrogel beads have been prepared by blending cellulose powder with chitosan solution, in the presence of ethylene glycol diglycidyl ether, as cross-linker. These hydrogel beads are denser, have improved chemical stability in the solutions with pH values down to 1, and have high adsorption capacity for Cu adsorption [99]. Cellulose/chitosan hydrogels have been also prepared when the polymers were mixed in NMMO and then processed under a pressure of 70 kg/cm² exerted by a compression molding machine, at 100 °C for 8 min. The films have smooth surface when containing 3 wt% chitosan, while for a higher concentration, of 5 wt% chitosan, they become rough due to phase separation [105]. These blended films have shown nondiffusible antibacterial properties.

Cellulose/xanthan hydrogels have been obtained from different allomorphic forms of cellulose (cellulose I, II, and III) and xanthan, by using epichlorohydrin as cross-linker [101]. The increase in xanthan concentration determined an increase in swelling degree of hydrogels, while using cellulose II led to the highest value of

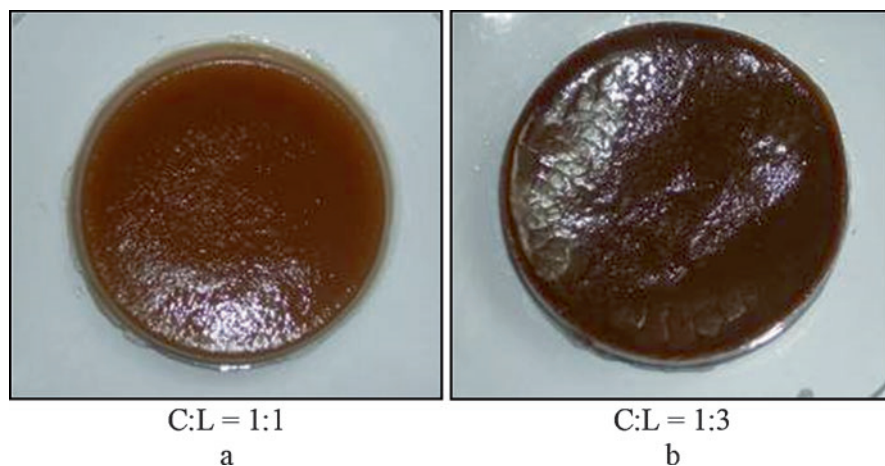


Fig. 7 Photo images of CL hydrogels with different compositions: (a) CL3, (b) CL5. (Reprinted with permission from [106]. Copyright © 2013, Cellulose Chemistry and Technology)

the swelling degree (2146%), which demonstrates the superabsorbent character of these hydrogels. The optical image of the hydrogel reveals a smooth surface and a homogeneous aspect, explained by a good incorporation of cellulose within the xanthan matrix.

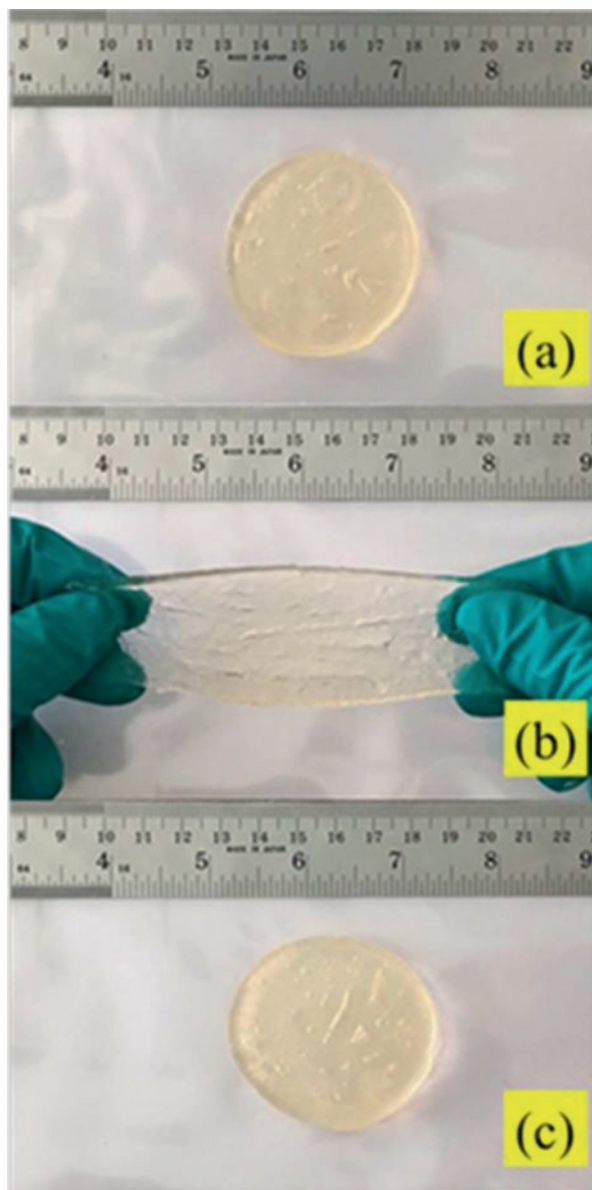
Cellulose/lignin hydrogels were prepared in the presence of epichlorohydrin [102]. The color of the hydrogels changes from light to dark brown, as the incorporated lignin amount increases. It appears that the structure is uniform and densely packed (Fig. 7) [106].

The swelling capacity of these hydrogels increased by increasing lignin content (from 2300 to 3000 wt%). This behavior is explained by the chemical uniqueness of lignin, which contains a high number of polar groups (aromatic and aliphatic hydroxyls, carbonyls, ethers). By involving lignin in cross-linking reactions, some hydroxyl groups are blocked by forming C-O-C linkages, but many other remain free, conferring hydrophilicity to the resulted 3D matrix; thus, pores with different structures and polarities could be formed. An increase in lignin content in cellulose/lignin hydrogels produces a more relaxed network with higher swelling capacity.

Cellulose/sodium alginate hydrogels have also been reported. The introduction of sodium alginate into cellulose hydrogels significantly increases the pore size and swelling ratio of hydrogel samples, while cellulose contributes to enhancing the mechanical properties of the hydrogels. These hydrogels have macroporous structure, excellent mechanical strength, and high equilibrium swelling ratio in water [103].

Bacterial cellulose/gelatin double-network (DN) hydrogels have been synthesized. It was observed that gelatin gel was brittle and easily broken into fragments under a modest compression, while the fracture strength and elastic modulus of a BC/gelatin DN gel were several orders of magnitude higher than those of gelatin gel and almost equivalent to those of articular cartilage. A similar enhancement on the

Fig. 8 Photograph of bacterial cellulose-based hydrogel composite (a) original form, (b) stretchable form, (c) reformable form. (Reprinted with permission from [19]. Copyright © 2017, Elsevier.)



mechanical strength was reported for the combination of bacterial cellulose with other polysaccharides such as sodium alginate, gellan gum, and carrageenan [107].

Bacterial cellulose and gelatin were successfully used to develop a hydrogel composite material [19]. The bacterial cellulose network was filled with gelatin, the glutaraldehyde was employed as a cross-linker, and a malleable and easily shaped at ambient temperature without any external stimuli hydrogel was obtained (Fig. 8).

The hydrogel composites presented excellent benefits in terms of their thermal stability, chemical resistance, and mechanical properties. Moreover, the swelling ratio of the hydrogel network in water was estimated to be 400–600%.

As many CNC-containing hydrogels target biological or biomedical applications, there has been an increasing amount of research on natural polymer-based hydrogels, primarily based on alginate [108–110], gelatin [111–113], and agarose [114, 115].

Cellulose can also be blended with *synthetic polymers*, such as poly(vinyl alcohol) [116, 117], poly(ethylene glycol) [118], and poly(N-isopropylacrylamide) [119], in order to obtain novel materials for special applications.

Cellulose/PVA hydrogels obtained by physical cross-linking method are of importance for biomedical applications, due to the avoiding of residual amounts of toxic chemical cross-linker and of a higher mechanical strength, higher than that obtained by chemical cross-linking method [116]. The effect of cross-linking methods on structure and properties of cellulose/PVA hydrogels revealed that the chemical cross-linked hydrogels, prepared in the presence of ECH, have a high swelling ratio but low mechanical strength as a result of the weak hydrogen bonding between cellulose and PVA. However, physical hydrogels prepared via solution blending of cellulose and PVA and repeating freezing/thawing cycles exhibit a dense structure between cellulose and PVA, leading to high mechanical strength.

Novel cellulose/PVA hydrogels obtained via physical cross-linking method showed an improved swelling capacity and an increased strength of the hydrogels, due to the presence of cellulose within the 3D network [117]. Moreover, the study regarding the release of vanillin from cellulose/PVA matrices demonstrated an increased percent of bioactive agent released with the increasing of cellulose in the hydrogels and also the shortening of the half time and maximum time of release.

In addition, BC/PVA hydrogels with mechanical properties similar to that of soft tissues have been successfully prepared [120]. Specifically, compositions of the hydrogels that have mechanical properties similar to that of the porcine aorta and the aortic heart valve have been identified. For hydrogel preparation, BC fibers produced by the *Acetobacter xylinum* with an average diameter of 50 nm have been used in combination with PVA and physically cross-linked through freezing/thawing cycles to form biocompatible nanocomposites with good mechanical properties.

CNC/PVA hydrogel showed moderate increases in mechanical properties toward the relatively weak PVA-only hydrogel [121–123]. CNF nanocomposite hydrogels have been used as absorbents due to their relatively high swelling capacity. The macroscopic core-shell hydrogels based on PVA and alginate showed higher storage and shear moduli accompanied by slightly increased dye absorption capacity when CNFs were chemically cross-linked into the core [124].

New hydrogels were prepared from cellulose and poly(N-isopropylacrylamide) (PNIPAAm) with high mechanical strength and adjustable thermosensitivity [119]. The first network was obtained by the chemical cross-linking of cellulose in NaOH/urea aqueous solution, in the presence of epichlorohydrin (ECH), while the second network was prepared by in situ polymerization/cross-linking of

N-isopropylacrylamide in cellulose network. The obtained hydrogels exhibit high mechanical strength and adjustable thermosensitivity, which strongly depend on the weight ratio between of both polymer networks.

Thermoresponsive poly(N-isopropylacrylamide) (PNIPAM) polymer networks have also been investigated with CNC fillers, showing characteristic temperature-dependent swelling properties along with tunable mechanics [79, 125]. Yang et al. further demonstrated how CNC surface charge and aspect ratio affected reinforcing ability in polyacrylamide hydrogels, showing that a higher surface charge density led to better dispersibility, while a higher aspect ratio contributed to increased mechanical properties [126].

Synthetic hydrogels like PEG-based hydrogels have advantages over natural hydrogels, such as the ability for photopolymerization, adjustable mechanical properties, and easy control of scaffold architecture and chemical compositions, but PEG hydrogels alone cannot provide an ideal environment to support cell adhesion and tissue formation due to their bio-inert nature [52, 127].

High-strength cellulose/poly(ethylene glycol) gels have been prepared by swelling cellulose gel membranes in a low-molecular-weight polyethylene glycol (PEG). The cellulose gel membranes have been fabricated by a pre-gelation method through employing cellulose solutions in aqueous NaOH/thiourea at low temperature [66]. These materials exhibit high mechanical performance, and their tensile strength increased sharply from 3.5 to 7.9 MPa with an increase in the molecular weight of PEG from 200 to 800 g/mol [52, 118].

Recently, the ionic interactions between amine-terminated PEG and carboxylated CNFs allow these nanocomposite hydrogels to exhibit a reversible sol-gel transition upon either strain or temperature ramping [79, 128].

6 Factors Affecting the Swelling Capacity of Cellulose-Based Hydrogels

There is little research on the swelling of physical gels due to their weak structure or low swelling, so the majority of the literature is related to the swelling of chemical hydrogels [54].

In order to investigate the *effect of the cross-linking degree on the swelling properties of cellulose hydrogel*, two types of swelling experiments were performed: (i) gels just after synthesis were placed in water, and their volume and weight at equilibrium, i.e., after cellulose complete coagulation, were measured, and (ii) cellulose cryogels swelling in water was recorded (Fig. 9) [59].

The influence of ECH concentration on the swelling degree of cellulose hydrogels (never dried, at equilibrium) showed a dramatic increase up to the maximal value of $Q = 3500\%$ for $R = 1$ and then a decrease at higher R values, reaching $Q \approx 1500\%$. The increase in swelling with the increase of cross-linker concentration is “against” the trends known for classical polymer gels for which the increase in cross-linking leads to the decrease of swelling. The higher the cellulose concentration, the lower the swelling degree of hydrogels, as expected (Fig. 9a). Another interesting point is

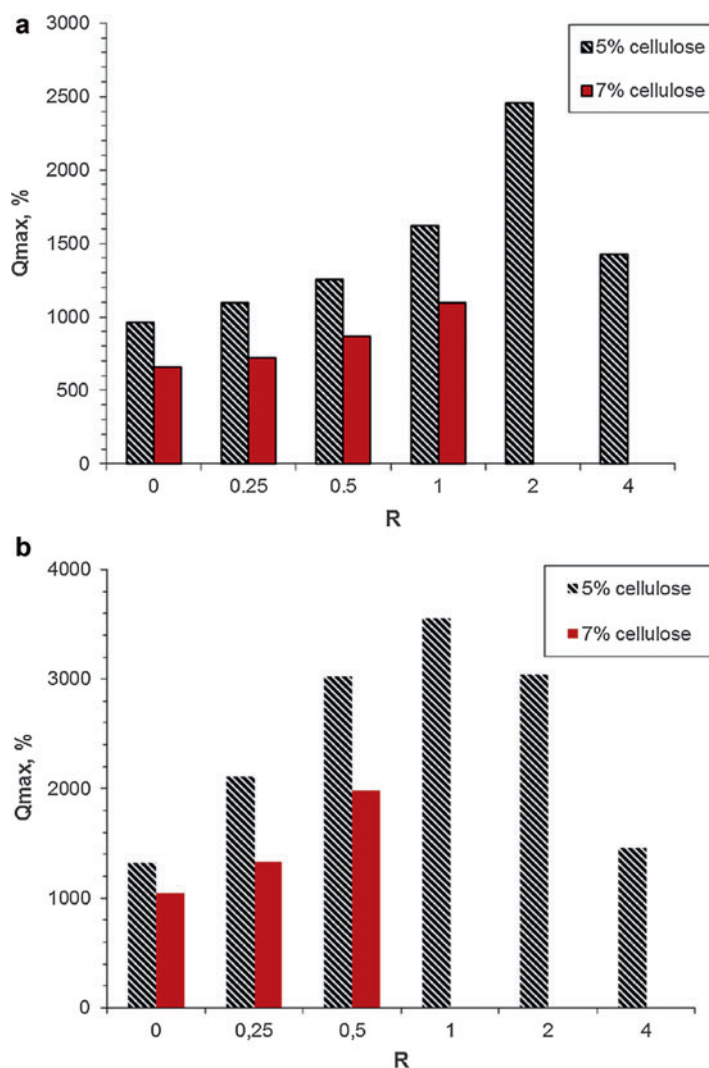


Fig. 9 Swelling degree of (a) cellulose hydrogels (never dried, at equilibrium) and (b) of cellulose cryogels in water at 37 °C, prepared from 5 to 7 wt% cellulose solutions, as a function of R. (Reprinted with permission from [54]. Copyright © 2017, Elsevier.)

that with the increase of ECH concentration, samples become more and more transparent: for example, samples prepared with $R = 0$ or $R = 0.25$ are opaque, whereas with $R \geq 2$ they are completely transparent. The transition from opaque to transparent coagulated swollen cellulose stands between $R = 1$ and 2. It may be deduced that the increase of cross-linking thus leads to a more homogeneous structure of coagulated cellulose [59].

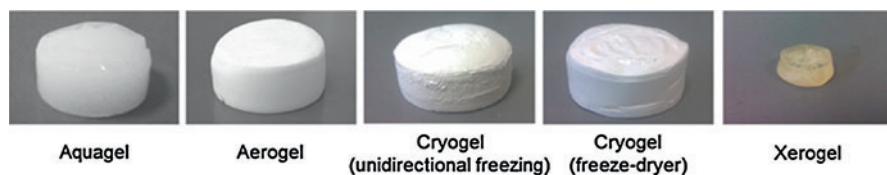


Fig. 10 Examples of an aquagel and samples after drying with different techniques (aerogel, cryogels, and xerogel) obtained from 7 wt% cellulose solution. (Reprinted with permission from [63]. Copyright © 2016, Springer International Publishing AG.)

Similar to hydrogels, the degree of swelling of cryogels in water first increases and then decreases with the increase of ECH concentration, for both studied cellulose concentrations (Fig. 10b). However, the absolute values of cryogel swelling are lower: for example, for samples prepared from 5 wt% solution, the minimal swelling of cryogel is around 900% and the maximal is around 2500%, while for hydrogels it is 1300% and 3500%, respectively. Lower swelling of dried compared to never-dried cellulose is due to the well-known hornification phenomenon. What is important is that even freeze-dried samples show the “anomalous” swelling, i.e., increase in swelling degree with the increase of ECH concentration, up to a certain value.

It is known that the drying procedure affects the water absorbency capacity of hydrogels. Supercritical or freeze-dried hydrogels retain the microporosity of the gel systems and thus significantly increase the swelling properties compared to air-dried, oven-dried, or vacuum-dried hydrogels, whose capillary retention decreases due to the recrystallization of the biopolymers in the gel systems [54].

However, the morphology of the porous material obtained after lyophilization or supercritical CO_2 drying is very different. While freeze-drying leads to the formation of large pores and channels from several microns to tens of microns, due to the growth of ice crystals, supercritical drying better preserves fine network structure leading to the formation of mesopores and small macropores which is reflected by specific surface area of several hundreds of m^2/g [63, 129–131].

For a systematic study of the *influence of drying method on the morphology and properties of porous cellulose materials*, the “aquagels” obtained after the coagulation in ethanol of cellulose solution in ionic liquid/DMSO mixture, three different drying methods were employed: supercritical CO_2 (aerogels), freeze-drying (cryogels), and vacuum drying (xerogels) (Fig. 10) [63]. Aerogels and both kinds of cryogels are white and opaque, whereas xerogels are yellowish, translucent, and much more shrunk.

Vacuum drying induces strong pore collapse resulting in samples with very low porosity, not-measurable specific surface area, and densities close to that of microcrystalline cellulose. Supercritical CO_2 drying yields nanostructured aerogels with bulk densities between 0.12 and 0.215 g/cm^3 and specific surface area up to 300 m^2/g . Freeze-drying gives highly macroporous cryogels with much lower specific surface areas (at the order of tens of m^2/g), but which can be as light as 0.05 g/cm^3 . The difference in the density of aerogels vs. cryogels is due to sample volume

contraction during drying step which is more pronounced for aerogels of low cellulose concentrations [63].

The morphology of aerogels is hierarchically structured with pore size varying from few tens of nanometers between the fibrils to few microns between the “hairy” beads. Cryogel morphology is a sheet-like cellulose network with large interconnected pores of several microns. Moreover, it is also possible to vary the pore size within one cryogel sample by applying a unidirectional freezing before freeze-drying. This allows decreasing pore size in the direction of temperature gradient, from lower to higher temperatures [63].

The pore sizes are affected by the biopolymer concentration, the cross-linking degree, and the drying processes. The pore sizes are generally determined by SEM after lyophilizing (i.e., freeze-drying) the hydrogels, and the resulting cryogels mostly have pore sizes within the average range of 3–8 μm , which decrease with increasing biopolymer concentration and cross-linking degree [54].

In order to establish the *influence of polymer concentration on the average pore size of the hydrogels*, one example that focused on cellulose/lignin (CL) hydrogels was used (Fig. 11) [102]. For CL hydrogel preparation, cellulose was dissolved in NaOH solution at low temperature, it was mixed with different amounts of lignin (CL1, 3:1; CL2, 2:1; CL3, 1:1; CL4, 1:2; CL5, 1:3; as gravimetric ratios), and it was cross-linked in the presence of epichlorohydrin. The SEM micrographs of hydrogels obtained from cellulose (C) and from cellulose/lignin are presented in Fig. 11.

The cross section of the cellulose hydrogels indicates a macroporous structure with an increasing average pore size by increasing lignin content, a fact which suggests that a higher quantity of water can be uptaken. This assumption is in good correlation with the data obtained for swelling degree (Q_{max} , from 1145% for C hydrogel to 3061% for CL5 hydrogel). An explanation for this behavior is the fact that the incorporation of a higher amount of lignin in the network, owing to a bulky and branched structure, contributes to an increase of ether linkages and implicitly to an increase of the swelling degree of the hydrogels.

Figure 12 clearly evidenced that the hydrogels with a high amount of lignin exhibit a broader pore size distribution than the ones with small quantity of lignin [106].

Thus, it can be concluded that the polymeric ratio significantly influences the average pore size of hydrogels, as well as pore size distribution. By increasing the amount of lignin in the cellulose matrix, the pore size distribution becomes wider, and the share of large pores increases. Such a suggestion confirmed the data obtained from the swelling measurements, showing that the increase of lignin in the CL hydrogels permits a higher water uptake.

Recently, the influence of cellulose concentration on average pore diameter of aero- and cryogels [63] has been studied. For both types of cryogels, the pore size decreases with increasing concentration. For example, the pore size in cellulose cryogel (freeze-dryer) decreases from almost 6 μm to around 1.5 μm when the concentration goes from 3 to 11 wt%; the same trend is observed for prefrozen samples. Overall, using unidirectional pre-freezing and/or varying cellulose concentration allows tuning pore sizes in cellulose cryogels. A similar result, i.e., the

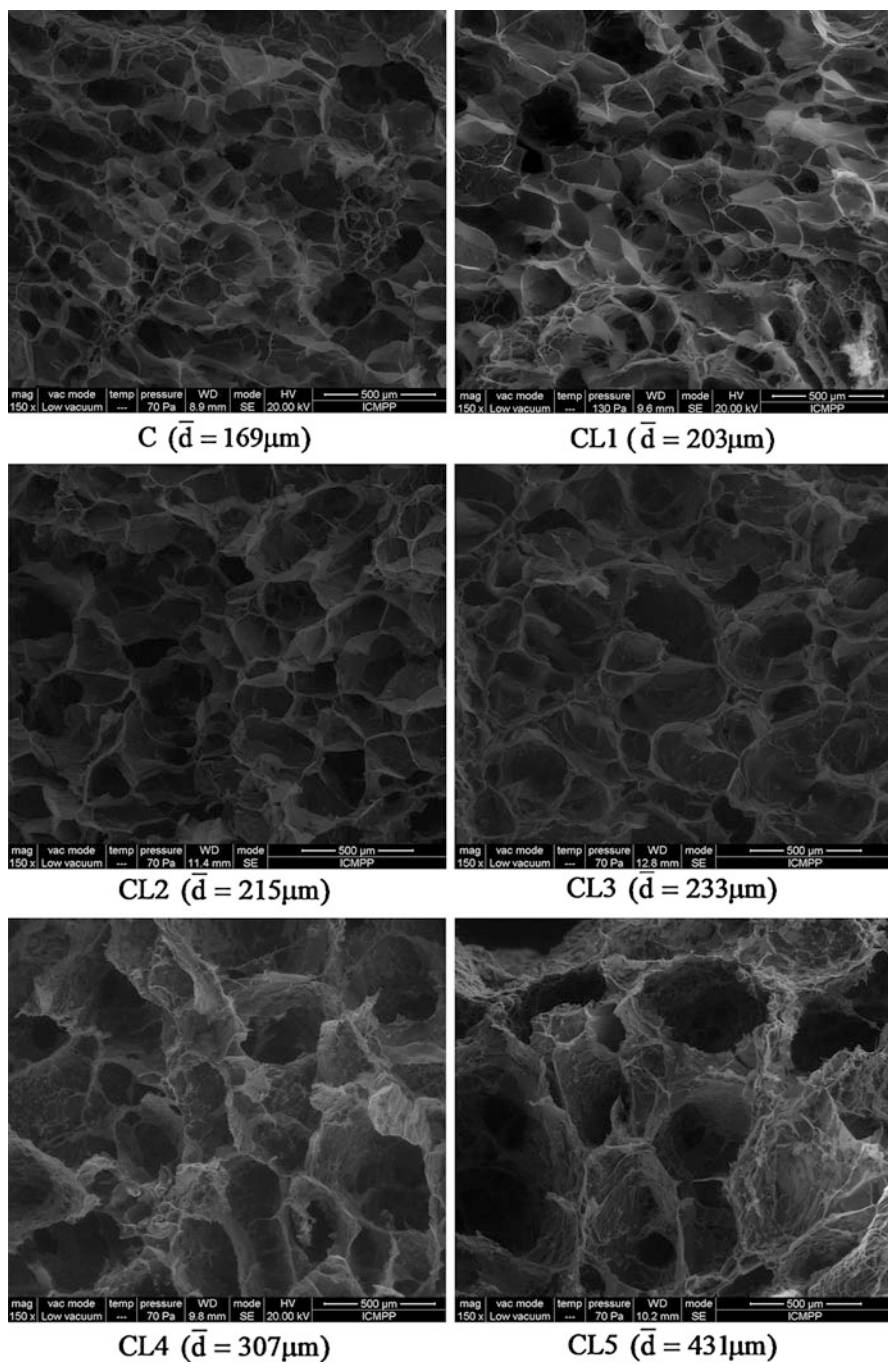


Fig. 11 SEM images of cellulose (C) and cellulose-lignin (CL) hydrogels. (Reprinted with permission from [102]. Copyright © 2012, Elsevier.)

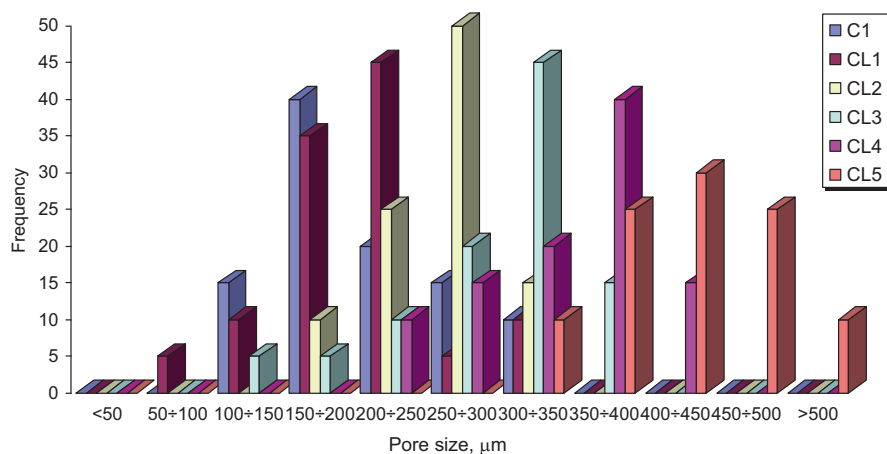


Fig. 12 Pore size distribution of cellulose-lignin hydrogels. (Reprinted with permission from [106]. Copyright © 2013, Cellulose Chemistry and Technology.)

decrease of pore size with the increase of cryogel density, was reported for microfibrillated cellulose [132]. Finally, xerogels are, due to strong pores contraction during slow vacuum drying, very dense, and their porosity is close to zero. No macro- or mesopores are visible on SEM images of xerogels [63]. As expected, vacuum drying cannot be used as a drying technique for the preparation of porous non-modified cellulose materials.

The swelling capacity of the hydrogels and their structural parameters, such as the degree of cross-linking, the average pore size, etc., are important parameters for their biomedical and pharmaceutical applications, i.e., the rate of drug release [133]. Depending on the network nature, chain dissolution may take place along with swelling due to the physical nature of the hydrogel network; thus, drug release results from the complex combination of swelling, diffusion, and erosion mechanisms [51]. When using hydrogels to modulate the drug release, the loading of the drug is performed either after cross-linking or simultaneously, during network formation [134]. Moreover, the bioactive molecule can be covalently or physically linked to the polymer network, to further tune the release rate.

Due to their large water content, hydrogels are highly biocompatible, possess rubbery mechanical properties close to those of soft tissues, and usually allow the incorporation of cells and bioactive molecules during the gelling [51, 135]. Moreover, although cells do not readily attach themselves to highly hydrophilic surfaces, the bulk or surface chemistry of hydrogels can be easily modified with extracellular matrix (ECM) domains, which promote cell adhesion as well as specific cell functions. Hydrogels are thus likely to be ideal platforms for the design of biomimetic scaffolds for tissue regeneration.

7 Applications of Cellulose-Based Hydrogels

Cellulose-based hydrogels are successfully applied to the fields as pharmacy, tissue engineering, and regenerative medicine, due to their unique properties such as hydrophilicity, biodegradability, soft and flexible nature, and biocompatibility. Therefore, cellulose and its derivatives have been used for drug delivery systems, as well as for burn wound regeneration and cardiac, vascular, neural, cartilage, and bone tissue regeneration:

- (a). **Cellulose I-Based Hydrogels:** Hydrogels based on bacterial cellulose mimic basic living processes and are of growing importance as bioactive scaffolds, such as skin tissue materials [136], meniscus implant [96], scaffold for chondrocyte proliferation [137], and dental [97]. In medical field, the main applications are wound dressings [138]. Nanoparticles from cellulose I are generated from bacterial cellulose and cellulose from plants. These cellulose nanoparticles were used in drug delivery systems [76, 139], or as scaffolds in artificial ligaments or tendon substitutes, with an excellent cytocompatibility [140]. The formulations from cellulose nanoparticles and silver nanoparticles can be used as microbial medicaments, antibacterial agents in wound dressing, bandages, implants, skins replacements for burnings, face masks, artificial blood vessels, cuffs for nerve surgery, drug delivery, cell carriers and support matrices for enzyme immobilization, cosmetic tissues, etc. [141].
- (b). **Cellulose II-Based Hydrogels:** The main areas in which cellulose II-based hydrogels are used as biomaterials are in contact lenses [142], wound coverings [143], drug delivery systems [59, 92, 102], and organ and tissue replacements, such as skin, tendon, cartilage, heart valve stents, and bone [54, 139].

8 Conclusions

This chapter emphasizes the recent advances in the design and synthesis of cellulose-based hydrogels. Different methods of cellulose dissolution in common organic solvents, or in novel solvents, such as LiCl/DMAc, NMMO, ILs, and NaOH/urea (or thiourea) aqueous solution, have been reported. A systematic investigation on the physical-chemical properties of the 3D cellulosic networks was done. In addition, the important factors that have an influence on the swelling capacity of cellulose-based hydrogels have been discussed. The large availability of cellulose in nature and its attractive advantages, such as biocompatibility, biodegradability, low cost, and non-toxicity, alongside the possibility to design various formulations of composite hydrogels, meet the demands for many areas of medical applications. A growing interest in the development of bio-nanocomposite structures based on cellulosic materials is expected. In the near future, cellulose-based hydrogels might be ideal platforms for the design of scaffold biomaterials in the field of tissue engineering and regenerative medicine.

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